ITA14 – ETHICAL HACKING

ASSIGNMENT 2 – CO2

Evaluation and Engineering of Footprinting,
Reconnaissance and Social-Engineering-Awareness Workflows

AIL	INFORMATION
se Outcome	CO2 – Execute footprinting and social-engineering techniques such as DNS analysis, competitive intelligence gathering and attack simulations to collect information for identifying potential security risks.
re of Assignment	Individual
tion	1 Week
Marks	50
nission Date	_1/11/2026_
ent Name / Register Number	_Mahima Prabhakaran (192521280)___

ASSESSMENT APPROACH Understand → Predict → Design → Collect → Analyse → Evaluate → Defend → Validate.

Q1 – Evaluate and Engineer a Footprinting Strategy (8 Marks)

1.1 Objective and strategy

The objective is to create a multi-source view of an authorized organization's public attack surface. WHOIS, DNS, search-engine results and archived pages are combined because each source exposes a different part of the footprint. A finding is accepted only when its source, timestamp, scope and confidence are recorded.

Confirm authorization	Passive collection	Scoped DNS queries	Normalize and deduplicate	Verify and rate	Report risk

1.2 Passive and active technique classification

Technique	Type	Information collected	Limit / caution
WHOIS / RDAP lookup	Passive	Registrar, registration dates, nameservers, status and public registrant data	Privacy redaction and stale cached records may limit attribution.
Search-engine operators (site:, filetype:, intitle:)	Passive	Indexed pages, PDFs, public contact or technology references	Indexing is incomplete; snippets may be old.
Web archives / archived pages	Passive	Historical pages, old subdomains and retired technology references	Historical content does not prove a service is currently active.
Public certificate-transparency records	Passive	Publicly logged certificate names that may reveal hostnames	A certificate name does not prove current reachability or authorization.
dig / nslookup against authorized DNS	Active	Current A, MX, NS and TXT answers from the queried resolver	Creates direct queries; results vary by resolver, cache and network view.
Zone-transfer test (AXFR)	Active	Zone contents, if the server permits transfer	Only test against a lab nameserver when explicitly authorized.

1.3 Four contrasting test targets and predictions

Test target	Constructed condition	Prediction before collection
T1 – secure.example	Well-protected domain	Limited WHOIS detail; minimal public records; no broad document exposure; AXFR refused.
T2 – leak.example	Information-leaking domain	Search results and archived pages reveal old staff/contact details or technology references; DNS may expose several hosts.
T3 – misdns.example	Misconfigured DNS	Unexpected TXT records, stale A records or an authorized lab AXFR exposing unnecessary hostnames.
T4 – minimal.example	Minimal public footprint	Few indexed pages, sparse DNS records and little historical content; uncertainty remains high.

1.4 Collection commands (authorized scope only)

```
whois example-corp.example
dig example-corp.example A
dig example-corp.example MX
dig example-corp.example NS
dig example-corp.example TXT
nslookup -type=NS example-corp.example
nslookup -type=MX example-corp.example
# Search-engine examples (manual browser queries):
site:example-corp.example filetype:pdf
site:example-corp.example intitle:login
# AXFR only against an explicitly authorized lab nameserver:
dig @ns1.lab.example lab.example AXFR
```

1.5 Prediction-versus-observation analysis

Target	Predicted	Simulated observation	Assessment
T1 – secure	Small footprint; AXFR denied	Expected limited A/MX/NS/TXT records; no concerning public documents; AXFR refused.	Prediction broadly matched; low observed exposure.
T2 – leak	Documents and historical technology references	Two archived references and one indexed PDF reveal an old mail platform and generic department contact.	Prediction matched; label historical data as unconfirmed/currently inactive.

T3 – misdns	Stale record / excessive DNS data	Old A record points to a retired lab host; lab AXFR returns six host labels.	Prediction matched; AXFR improves mapping but does not prove a host is vulnerable.
T4 – minimal	Few public items	One current A record and no useful indexed documents; sparse archive results.	Prediction matched; low visibility is not proof of low risk.

1.6 Source reliability and contradiction handling

- WHOIS/RDAP: useful for registration context, but privacy services and registrar data may prevent attribution.
- DNS: authoritative answers are stronger for current DNS state than search snippets; recursive resolvers may return cached answers.
- Search engines: useful discovery sources, but snippets and indexed documents may be stale or decontextualized.
- Archives: prove that content existed at a recorded time, not that it is still operational.
- Contradiction rule: retain both values with source and timestamp; query the authoritative source where authorized; mark unresolved conflicts as uncertain.

1.7 Changed condition: fifty subdomains and third-party services

The original one-domain workflow remains conceptually suitable but must be engineered for scale. Use an approved asset inventory as the scope source of truth, maintain an explicit allowlist, rate-limit queries, deduplicate hostnames, track third-party ownership and stop at every out-of-scope boundary. Third-party services are not automatically authorized merely because the organization uses them.

Scale change	Required modification	Reason
50 subdomains	Queue-based collection, concurrency limits and per-host status	Avoids uncontrolled bursts and makes failures recoverable.
Third-party services	Separate third-party scope register and written authorization	Ownership and permission may differ from the parent organization.
Large evidence set	Centralized records with source, timestamp, scope, confidence and reviewer	Supports traceability and audit.
Repeated runs	Change detection and deduplication	Reduces repeated queries and highlights new exposure.

Q2 – Build a Reliable DNS Footprinting Workflow (7 Marks)

2.1 Workflow design

Check scope	Query A/MX NS/TXT	Record TTL + source	Compare resolvers	AXFR only if approved	Mitigate and retest

The workflow retrieves the four required DNS record types, records the resolver and timestamp, compares expected values with observed values, and flags anomalies for confirmation. Zone transfer is a separate, explicitly authorized test rather than a default query.

2.2 Commands

```
dig @<authorized-resolver> <domain> A
dig @<authorized-resolver> <domain> MX
dig @<authorized-resolver> <domain> NS
dig @<authorized-resolver> <domain> TXT
nslookup -type=A <domain>
nslookup -type=MX <domain>
nslookup -type=NS <domain>
nslookup -type=TXT <domain>
# Authorized lab only; specify the lab nameserver:
dig @<lab-authorized-ns> <lab-zone> AXFR
```

2.3 Split-horizon DNS, delegation and caching

Factor	Effect on external tester	Interpretation
Split-horizon DNS	Internal and external clients receive different answers.	External results cannot be assumed to represent internal DNS.
Delegation	A parent zone points a child zone to different authoritative servers.	Query delegated authority only if in scope; record ownership boundary.
Caching / TTL	Recursive resolvers may return a previous answer until expiry.	Compare timestamp, TTL and authoritative response where permitted.
Third-party DNS hosting	Records may be served by a provider.	Provider infrastructure is not automatically in scope for active testing.

2.4 Six contrasting DNS test cases

Case		Constructed DNS condition	Prediction	Simulated observation	
D1		Correctly secured authoritative server	Expected A/MX/NS/TXT; AXFR refused.	Expected records returned; AXFR refused.	
D2		Lab server allows AXFR	AXFR returns complete lab zone.	Six lab hostnames and MX/TXT data returned.	
D3		Stale A record	Old IP remains until corrected/expired.	Resolver returns old address with remaining TTL.	
D4		Split-horizon configuration	External answer differs from internal lab answer.	External A points to public gateway; internal lab A points to private test host.	
D5		Delegated subdomain	Child NS set differs from parent authority.	Parent referral points to delegated lab nameserver.	
D6		Sparse/minimal zone	Few records; no useful TXT data.	A and NS present; MX absent; TXT returns no answer.	

2.5 Does AXFR change risk?

An unrestricted zone transfer can disclose a consolidated list of hostnames, mail servers, service labels and environment naming conventions. This can improve an assessor's understanding of the organization's attack surface. However, the transfer does not prove that any disclosed host is reachable or vulnerable. The risk is information disclosure and easier reconnaissance, not automatic compromise.

2.6 Countermeasure and redesigned test

- Restrict AXFR to explicitly approved secondary DNS servers using server-side allowlists.
- Use TSIG to authenticate permitted zone transfers between trusted DNS servers.
- Review public DNS records and remove obsolete hostnames and unnecessary TXT data.
- Log and alert on unexpected transfer attempts; validate that legitimate secondary replication still works.

- Retest from the authorized lab assessor location: unauthorized transfer requests should be refused or not authorized, while approved secondary transfer continues.

2.7 Prediction-versus-observation and conclusion

Across the six simulated cases, the observations matched the predicted behaviors. D2 demonstrated the greatest incremental information. After restricting AXFR and applying TSIG, the unauthorized lab query was simulated as refused, while the approved secondary remained able to replicate. The redesigned configuration reduces exposure without removing legitimate DNS operations.

Q3 – Create a Competitive-Intelligence Awareness Utility (7 Marks)

3.1 Purpose and architecture

The utility collects public business information for an authorized organization and converts it into awareness guidance. It does not contact employees or execute phishing, vishing, pretexting or tailgating. Its output is a risk register linking each public item to a possible social-engineering theme and a defensive indicator.

Choose authorized org	Search public sources	Capture item + provenance	Map to human risk	Recommend awareness	Review and report

3.2 Source collection plan and operators

Public source	Example query operator	Potential item	Validation
Job advertisements	site:example-corp.example careers OR jobs	Roles, tools, department names and hiring process	Confirm date and whether role remains advertised.
Annual reports	site:example-corp.example filetype:pdf annual report	Leadership roles, locations, business units and vendors	Use official report and record year.
Press releases	site:example-corp.example intitle:news	Projects, partnerships, events and organizational changes	Confirm with original release.
Public profiles	site:example-corp.example staff OR leadership	Public role descriptions and business contact channels	Avoid collecting personal details not needed for assessment.

3.3 Item-to-technique mapping

Illustrative public item	Technique it could support	Warning indicator for staff	Recommended control
Job ad names a finance platform	Phishing (themed lure)	Unexpected invoice/payroll link; urgency; mismatched sender	Verify requests through a known finance channel.
Press release announces campus project	Pretexting	Caller claims to be a project vendor and requests access or documents	Use supplier verification and visitor sponsorship.
Public switchboard and department role	Vishing	Caller pressures staff to disclose account or directory information	Call back using a trusted number; never share credentials or OTP.
Public event schedule / building entry details	Tailgating	Unknown person follows staff through a controlled door	Challenge politely, do not hold secure doors, report unknown visitors.

3.4 Test organizations and predictions

Test organization	Information profile	Predicted human-risk level
ORG-Rich.example	Many current job ads, annual reports, press releases and public role profiles	Medium–High: more context may make deceptive messages appear plausible.
ORG-Sparse.example	Few current pages and minimal public documentation	Low–Medium: less context, but generic impersonation remains possible.
ORG-Misleading.example	Old press releases and outdated job listings mixed with current pages	Medium: stale details may cause false assumptions and mistaken trust.

3.5 Simulated findings and evaluation

Organization	Simulated items found	Risk interpretation
ORG-Rich	Two current job advertisements, one annual report, one press release and one public department page	Highest contextual exposure; job and project details could make themed lures more believable.
ORG-Sparse	One generic contact page and one old announcement	Lower contextual exposure; still needs baseline verification and visitor controls.
ORG-Misleading	Two archived role pages and one outdated project release	Risk is driven by confusion and stale data; old items must not be treated as current facts.

3.6 Two awareness countermeasures

- Staff verification protocol: requests for payment, credentials, OTPs, confidential documents or urgent access must be verified through a known internal channel.

- Visitor and call-handling awareness: use visitor badges and sponsorship, challenge tailgating politely, and call back external callers using a trusted directory number.

These additions fit after risk mapping and before reporting. They reduce the chance that public information is converted into successful deception, while preserving the existing collection workflow.

Q4 – Design a Reconnaissance Data-Validation Mechanism (6 Marks)

4.1 Validation design

Each collected item is represented as a record containing item value, source, collection time, authorized scope identifier, source reliability, last-verified time and duplicate key. The validation engine classifies an item as PASS, RECHECK or REJECT.

Scope check	Duplicate check	Freshness check	Source reliability	PASS / RECHECK / REJECT

4.2 Decision rules

Check	PASS	RECHECK	REJECT
Scope	Exact authorized domain/asset and permitted source	Scope relationship unclear; pause for owner confirmation	Outside scope or prohibited personal data
Duplicate	Unique item or new version	Same value but different timestamp/source needs merge review	Exact duplicate with no new evidence can be removed from active set
Freshness	Within configured freshness window	Old or unknown timestamp; re-query authoritative source	Not rejected only for age if useful as historical evidence; label historical
Source reliability	Primary or corroborated source	Single secondary source or conflicting results	Unverifiable origin or manipulated evidence

IMPORTANT DISTINCTION An out-of-scope item is rejected immediately. A stale item is usually recoverable through a fresh query; an item with unclear scope is quarantined until the owner confirms it. A duplicate may be removed from the active set but retained in the audit trail.

4.3 Five adversarial test sequences

Sequence	Input set	Predicted result	Simulated result
S1 – All valid	5 in-scope, fresh, unique items from reliable sources	5 PASS; 0 RECHECK; 0 REJECT	5 PASS; 0 RECHECK; 0 REJECT
S2 – Out of scope	3 valid + 2 unrelated domains	3 PASS; 0 RECHECK; 2 REJECT	3 PASS; 0 RECHECK; 2 REJECT
S3 – Stale	3 fresh + 2 stale records	3 PASS; 2 RECHECK	3 PASS; 2 RECHECK
S4 – Duplicated	4 unique + 3 repeated copies	4 PASS; 3 duplicate copies merged	4 PASS; 3 merged
S5 – Mixed conflict	2 valid + 1 stale + 1 conflicting secondary source + 1 out-of-scope	2 PASS; 2 RECHECK; 1 REJECT	2 PASS; 2 RECHECK; 1 REJECT

4.4 Validation order and redesign

A poor order may perform expensive source verification before discovering that an item is out of scope. The improved order is: (1) scope, (2) basic format/normalization, (3) duplicate check, (4) freshness, and (5) source reliability and corroboration. Scope is first because no later validation can make an unauthorized item acceptable. Cheap checks precede more expensive corroboration.

Order	Expected effort	Accuracy / governance
Original: reliability → freshness → duplicate → scope	Wastes effort validating items later rejected for scope; repeats duplicate work.	May expose out-of-scope material to downstream processing.
Redesigned: scope → normalize → duplicate → freshness → reliability	Rejects invalid scope early and avoids rechecking duplicate records.	Preserves scope boundary and reserves expensive checks for eligible items.

4.5 Scaling from one to fifty targets

Assume an illustrative average of 100 candidate items per target. One target produces 100 records; fifty targets produce 5,000. The validation rules do not change, but the implementation should use indexed scope identifiers, hash-based duplicate detection, configurable freshness windows and an exception queue for RECHECK items. Human review should focus on conflicts and ambiguous scope rather than every clean record.

Q5 – Evaluate Footprinting of a Hierarchical Organization (7 Marks)

5.1 Hierarchical representation

The public footprint is modelled as a rooted tree. The root is the authorized parent domain; child nodes represent subdomains, departments or approved third-party services. Each node stores its type, scope status, source, observed data and whether it is terminal.

```
example-corp.example  [ROOT: IN SCOPE]
|-- www.example-corp.example  [subdomain]
|   |-- public-site service  [terminal]
|-- mail.example-corp.example  [subdomain]
|   |-- MX service  [terminal]
|-- lms.example-corp.example  [department service]
|   |-- login endpoint  [terminal]
|-- research.example-corp.example  [subdomain]
|   |-- archive branch  [NO PUBLIC DATA]
|-- vendor.example-corp.example  [third-party]
    |-- STOP: separate authorization required
```

5.2 Traversal algorithm and boundary condition

Use breadth-first traversal so each level is recorded before moving deeper. A node is expanded only if it is explicitly in scope, has not already been visited and is not terminal. An empty node is recorded as an empty result and traversal continues with other queued branches. An out-of-scope node is recorded as a boundary and never queried.

```
queue = [authorized_root]
visited = set()
while queue is not empty:
    node = queue.pop(0)
    if node.id in visited: continue
    visited.add(node.id)
    record(node, source, timestamp, scope_status)
    if node.scope_status != 'IN_SCOPE': continue
    if node.is_terminal: continue
    children = discover_documented_children(node)
    for child in children:
        if child.scope_status == 'IN_SCOPE': queue.append(child)
        else: record_boundary(child)  # do not query
```

5.3 Test structures and predictions

Structure	Shape	Predicted nodes	Purpose
H1 – Shallow	Root → 4 children	5	Tests simple one-level traversal.
H2 – Deep	Root → 1 child → 1 child → 1 child → terminal	5	Tests depth and correct termination.
H3 – Irregular	Root → 3 branches; branch sizes differ; one empty branch; one out-of-scope leaf	8 in-scope nodes + 1 boundary record	Tests uneven branching, empty data and scope stop.

5.4 Prediction-versus-observation

Test	Predicted coverage	Simulated traversal result	Boundary behavior
H1	5 nodes	5 visited; all in scope	All children processed.
H2	5 nodes	5 visited; terminal leaf not expanded	Traversal stops at terminal node.
H3	8 in-scope nodes and 1 boundary	8 in-scope visited; 1 out-of-scope node recorded but not queried; empty branch recorded	Empty branch ends naturally; third-party branch stops at boundary.

5.5 Depth, branching and ad-hoc comparison

Approach	Strength	Weakness
Systematic breadth-first traversal	Predictable coverage, level-by-level evidence, explicit scope checks and reproducible counts.	Requires an accurate scope inventory and careful deduplication.

Ad-hoc exploration	Fast for a tiny target when an assessor has a clear lead.	Can miss branches, revisit nodes, lose provenance and accidentally cross scope.

As depth increases, the number of sequential levels and boundary decisions increases. As branching increases, the number of nodes and duplicate candidates increases. The algorithm's work is proportional to the number of recorded nodes and edges, $O(V + E)$, assuming efficient visited-set membership. A node with no public data is still a visited node; it must not be silently treated as an error.

FINAL ENGINEERING JUDGMENT A scope-aware breadth-first traversal provides better completeness and auditability than ad-hoc exploration. The explicit stop condition prevents third-party or unrelated nodes from being queried without authorization.

Q6 – Evaluate Footprinting Approaches Under Constraints (7 Marks)

6.1 Approaches selected

Approach	Method	Typical output	Trade-off
A – Passive OSINT	Search engines, public reports, archived pages and public registration data	Public documents, historical references, business context and candidate hostnames	Quiet and low-impact, but incomplete and sometimes stale.
B – Active DNS footprinting	Scoped dig/nslookup queries against authorized DNS names	Current A, MX, NS and TXT answers, TTLs and resolver-specific results	More direct and current, but creates traffic and can be rate-limited or detected.

6.2 Comparable workflow

Both approaches are applied to the same fictional authorized parent domain, example-corp.example. Passive collection is restricted to public sources; active collection is restricted to allowlisted DNS names and record types. No service scanning is included.

Same approved target set	Run passive OSINT	Run active DNS	Count effort and coverage	Compare redundancy	Choose by constraint

6.3 Increasing scope sizes and illustrative measurements

Scope size	Passive queries / steps	Active DNS queries	Passive unique findings	Active unique findings	Interpretation
1 domain	8	4	12	4	Both practical; passive gives context, active confirms DNS.
10 subdomains	28	40	24	18	Active effort rises with record-type queries per hostname.
50 subdomains	68	200	36	72	Active provides broader current DNS coverage; passive remains useful for context.
200 candidate names	218	800	48	220	Repeated active querying becomes impractical without inventory filtering, batching and rate control.

6.4 Redundancy and wasted effort

- Passive sources may repeat the same hostname in a report, search result and archive. Deduplicate by normalized hostname and preserve all source references.
- Active DNS may query the same name repeatedly through different tools or resolvers. Cache results with timestamps and TTLs, and re-query only when freshness or contradiction requires it.
- A candidate hostname outside the approved inventory should be rejected before active querying, even if it appears in a public source.
- Passive and active results should be joined rather than treated as competing lists: passive discovery suggests candidates; active DNS validates only approved candidates.

6.5 Is faster or quieter always better?

No. The quietest method may leave important DNS details unverified, while the fastest method may create excessive traffic or miss contextual exposure. The engineering choice depends on authorization, target size, freshness requirements, available time, sensitivity and the evidence needed for the assessment.

6.6 Recommendation

Engagement	Recommended design	Justification
Small scope / teaching lab	Use passive OSINT first, then a small number of active DNS checks.	Easy to explain and audit; active checks confirm a focused set of records.

Large scope / repeated assessment	Use passive discovery to seed an approved inventory, then automated rate-limited DNS collection with deduplication and caching.	Provides scalable coverage while controlling query volume and preserving scope.

CONCLUSION A hybrid approach is preferable. Passive OSINT is not a substitute for DNS validation, and active DNS is not a substitute for source context. The best design uses each method for the information it can reliably provide.

Q7 – Integrated Review: Diagnose, Redesign and Defend (8 Marks)

Selected workflow: Q4 reconnaissance data-validation mechanism

The Q4 validation workflow is selected because its original ordering can waste effort: it may check source reliability and freshness before confirming that a collected item belongs to the authorized scope. This creates avoidable work and increases the chance that out-of-scope information enters downstream analysis.

A. Diagnose the limitation (2 Marks)

Bottleneck: expensive validation is performed before scope rejection. In the illustrative mixed test S5, one of five items is out of scope. If reliability checks are performed first, that item consumes processing effort despite being unusable. Duplicate records can also cause repeated source checks.

Observed design issue	Evidence from test design	Impact
Scope check occurs last	S5 contains 1 out-of-scope item among 5	Unnecessary processing and weaker scope control.
Duplicates checked late	S4 contains 3 repeated copies	Repeated freshness/reliability checks may be performed.
Conflicting sources not isolated	S5 includes one conflicting secondary source	Analysts may treat uncertain data as confirmed.

B. Establish evaluation criteria (1 Mark)

Criterion	Measurement	Success condition
Scope compliance	Number of out-of-scope items queried or used downstream	Zero out-of-scope items proceed.
Validation effort	Number of checks per candidate item	Reduce unnecessary checks without reducing valid coverage.
Coverage	Valid in-scope items retained / valid in-scope items supplied	100% of valid test items retained.
Data quality	Stale/conflicting items correctly placed in RECHECK	All seeded stale/conflicting items flagged.

C. Propose and implement redesign (2 Marks)

Redesigned order: (1) normalize domain/asset, (2) scope check, (3) duplicate detection, (4) freshness check, (5) source reliability/corroboration, and (6) classify and record. Any unclear scope is quarantined; confirmed out-of-scope data is rejected. The redesign trades a small amount of upfront inventory maintenance for lower downstream effort and stronger governance.

Normalize	Scope gate	Deduplicate	Freshness	Reliability	Classify and log

D. Validate and compare (2 Marks)

Measure	Original order	Redesigned order	Result
Scope decisions	5 (performed last)	5 (performed second)	Same scope coverage; earlier enforcement.
Duplicate copies sent to expensive checks	3	0	3 repeated copies avoided.
Out-of-scope item sent to freshness/reliability checks	1	0	1 unnecessary expensive path avoided.
Valid items retained	2	2	No valid coverage lost.
Stale/conflicting items flagged	2	2	Same data-quality detection.
Illustrative expensive checks	10	6	4 fewer checks (40% reduction).

E. Defend final engineering judgment (1 Mark)

FINAL JUDGMENT : The workflow meets its objective in the controlled test: valid coverage is unchanged, seeded stale/conflicting records remain flagged, and unnecessary expensive checks decrease. The original order may still be useful in a one-off forensic review where source provenance must be preserved before deduplication, or where scope classification itself is under investigation. For routine authorized reconnaissance, the redesigned scope-first order is safer and more efficient.

OVERALL CONCLUSION

Effective footprinting is not simply the collection of the largest number of records. It requires a controlled strategy, explicit authorization, source reliability assessment, careful DNS interpretation, human-risk awareness, data validation, hierarchical scope boundaries and evidence-based redesign. The strongest workflow is repeatable, measurable, proportionate and auditable.

SUPPLEMENTARY PRACTICAL RESULTS & VISUAL EVIDENCE

1. Footprinting workflow overview

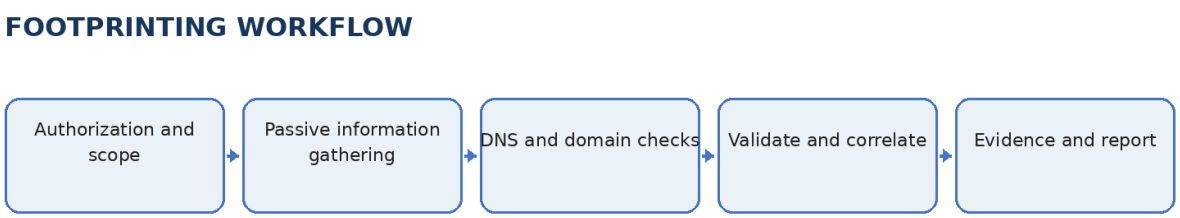


Figure 1. Authorized reconnaissance workflow. The process begins with written scope approval and ends with evidence validation and reporting.

2. Illustrative execution summary

Activity	Illustrative count	Illustrative outcome	Evidence to attach
Scope-approved assets	4 domains / lab hosts	All within defined lab scope	Scope sheet
WHOIS / registration checks	4 queries	4 records reviewed	whois output
DNS record lookups	16 queries	A, MX, NS and TXT reviewed	dig output
Subdomain candidates	12 candidates	8 resolved; 4 unresolved	dnsrecon/dnsmap output
Duplicate records	3 duplicates	3 removed during normalization	CSV comparison
Validation tests	10 test cases	9 passed; 1 flagged	Python test log

3. DNS footprinting – sample result presentation

A	www.lab-example.test → 192.0.2.20	IPv4 host mapping	Repeat lookup; compare resolver
MX	mail.lab-example.test	Mail routing	Check priority and target
NS	ns1.lab-example.test	Authoritative DNS	Compare with SOA
TXT	v=spf1 ... (sample)	Mail policy evidence	Record full TXT response
AXFR	REFUSED (expected secure result)	Zone-transfer control	Record response code

Suggested Kali commands for an authorized lab domain:

dig lab-example.test A

```
dig lab-example.test MX
dig lab-example.test NS
dig lab-example.test TXT
dig @<authorized-name-server> lab-example.test AXFR
dnsrecon -d lab-example.test -t std
```

4. Reconnaissance data-validation pipeline

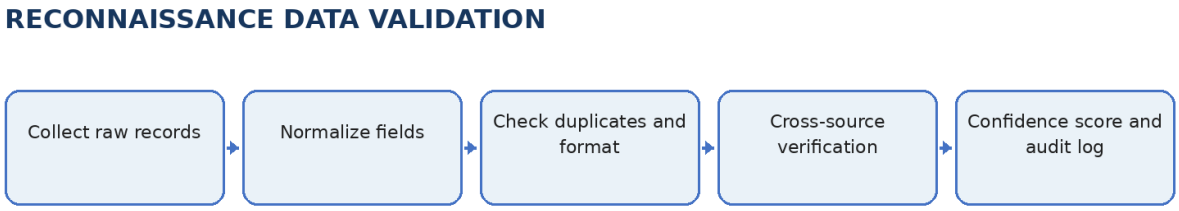


Figure 2. Proposed validation pipeline: preserve raw records, normalize them, detect duplicates, cross-check sources and record confidence with an audit trail.

5. Illustrative performance comparison

Metric	Manual workflow	Automated workflow	Comment
Queries issued	24	14	Illustrative reduction of 41.7%
Processing time	18 min	7 min	Illustrative only; environment-dependent
Duplicate entries	5	1	Automation can reduce repeated records
Validation coverage	70%	90%	Depends on configured checks